Size rule and inherited-f4 fold, end to end (2026-11-12)

Size rule `fes-then-inherited` and the `inherited-f4` fold, end to end (2026-11-12)

Owner's ask (2026-11-12): "do the rest of this stuff. Keep working on these things", after the previous round (PR #3979: `../m4gb_update_signatures_20261112/`, `../inherited_f4_next_20261112/`). This study lands two follow-ups that round named.

What changed (`src/cryptanalysis/ic_framework/solvers.rs`)

1. **Fold.** The engine `inherited-f4` now runs with the next kernels (SIMD row XOR, fused tallies, Four Russians table reuse, exact fused builders): the same rows, decisions and word-XOR count, measured at 0.86 to 0.94× the wall per call in `../inherited_f4_next_20261112/` (run 1). The previous kernels stay runnable as `inherited-f4-legacy`, the A/B arm. `inherited-f4-next` stays registered as an alias of `inherited-f4`, so the earlier sweeps reproduce.
2. **Size rule.** A new solver `fes-then-inherited` runs `fes-f2` on quadratic systems of at most `fes_max_vars` unknowns (default 14, at most 64 equations) and `inherited-f4` on everything else. The cost record says which engine ran (`size_rule_fes`). The default 14 was fixed from the per-call word-XOR counts before this run: at 14 unknowns `fes-f2` counts about 33 thousand against `inherited-f4`'s 49 to 52 thousand; at 16 unknowns 132 thousand against 102 to 124 thousand (`../inherited_f4_next_20261112/RESULTS_AB.md`). Both engines count word XORs, so S stays on one pricing basis.

Test: `ic_framework::solvers::tests::size_rule_and_inherited_fold_match_the_reference` (both branches of the rule against the exhaustive reference; folded and legacy `inherited-f4` give the same solutions and count). The module's 14 tests pass.

Protocol

`run_e2e.sh` under `tools/benchrec.py run`: `ic bench --sweep` on the six instances of the e2e ladder (Koblitz 13, 17, 19, 23; random 15, 21; same subspace bases), the algebraic oracle with `inherited-f4`, `inherited-f4-legacy` and `fes-then-inherited`, the pair table as the reference row, 3 repeats, 60 s per call, 16 matched rho runs. Local container, one thread, not isolated; no leaderboard row.

Pre-stated expectation and falsifier

- `inherited-f4` and `inherited-f4-legacy`: identical S on every instance (same count), the folded engine with the lower wall per call.
- `fes-then-inherited`: S equal to `fes-f2`'s at degree 13 (14 unknowns: the rule picks `fes-f2`, which gave 94.7 against 123.1 in the previous run), and equal to `inherited-f4`'s on the other five instances (16 unknowns and up: the rule picks `inherited-f4`). So the size rule's row is the best algebraic row on every instance where the two engines differ.
- Falsifier: folded and legacy S differ on any instance; the rule's S at degree 13 not below `inherited-f4`'s; or the rule's S on any other instance differing from `inherited-f4`'s beyond run noise.

Results `[pcc:unverified]`

Run 2026-11-12 16:12 to 16:16 Z, record `benchmarks/records/20261112T1612Z-size-rule-fold-e2e.json` (measured; raw `results.tar.gz`, `results.json`), binary SHA-256 in `results/ic-sha256.txt` (a frozen copy of commit 4840612771's build). Full table `RESULTS.md`, graph `results.svg/png`, `REPORT.pdf`. Every one of the 24 configurations recovered its planted logarithm in every repeat, and so did every matched rho run.

instance (unknowns)	rho S	pair table S	`inherited-f4` S	`-legacy` S	`fes-then-inherited` S (engine it ran)
Koblitz 13 (14)	2.89	197.8	138.0 (117.8–167.6)	138.0	**101.9 (83.7–116.3)** (`fes-f2`)
random 15 (16)	2.28	473.7	278.8	278.8	278.8 (`inherited-f4`)
Koblitz 17 (18)	1.73	493.2	304.5	304.5	304.5 (`inherited-f4`)
Koblitz 19 (20)	1.35	1,063.3	465.5	465.5	465.5 (`inherited-f4`)

instance (unknowns)	rho S	pair table S	`inherited-f4` S	`-legacy` S	`fes-then-inherited` S (engine it ran)
random 21 (22)	1.24	1,975.8	4,763.3	4,763.3	4,763.3 (`inherited-f4`)
Koblitz 23 (24)	0.88	3,986.3	2,051.2	2,051.2	2,051.2 (`inherited-f4`)

Mean over 3 repeats (range where it matters).

- **The fold changes no S**: `inherited-f4` and `inherited-f4-legacy` agree to every digit on all six instances, as the identical word-XOR counts require.
- **The size rule is the best algebraic row on every instance.** At degree 13 it runs `fes-f2` and gives S 101.9 against `inherited-f4`'s 138.0 (0.74×; 35.3× the matched rho instead of 47.8×), the three repeats not overlapping (83.7–116.3 against 117.8–167.6). On the other five instances it runs `inherited-f4` and has exactly its S. The pair table stays best on random
  21. The expectation held; the falsifier did not fire.
- **The degree-13 figures differ from the previous run** (fes-f2 94.7 and inherited-f4 123.1 over 2 repeats): the third repeat of this run is the dearer one for both engines (116.3 and 167.6), and the matched rho mean moved from 2.76 to 2.89 with its own random runs. The ratio is 0.77× there and 0.74× here.
- **Wall inside the pipeline is not a paired comparison.** The sweep runs the configurations one after another on shared per-thread caches, so the folded `inherited-f4` (run first) is below `-legacy` per call on four instances (0.89 to 0.92×) and above it at Koblitz 13 and 17 (1.05 and 1.08×). The paired per-call measurement of the same kernels is `../inherited_f4_next_20261112/` run 1 (0.86 to 0.94× on every cell).

No leaderboard row: no replay certificate or validator report (validation gate); a validated isolab run of the degree-13 configuration is being prepared separately (`../fes_k13_validated_20261112/`).

Sources checked

KB (`research-kb`) unavailable in this session.
`src/cryptanalysis/ic_framework/mod.rs` (solver pricing: word-XOR units priced by count), `src/cryptanalysis/ic_framework/solvers.rs`, `research/inherited_f4_next_20261112/README.md` and `RESULTS_AB.md`.

Per instance: the matched counted rho (mean S over its runs) and every configuration run end to end. `S` = total group-addition equivalents / sqrt(r), mean over repeats (min–max); `S / rho` = mean s_over_rho; `ok` = every repeat recovered the planted logarithm; `hit` = decomposition hit rate; `tried` = targets tried; `solver` = engine, priced_by, mean calls, budget hits, share of the row's GAE, wall per call.

instance (r)	configuration	S mean (min–max)	S / rho
icv1-f2m15-tm249-9e24ae34 (r = 5503)	**matched rho** negation-map r-adding walk: look-ahead, short-cycle escape by doubling, distinguished points, stride starts, 16 runs	2.28	1.00
	binary-subspace[dimension=8] + mitm[m=2] + – + walk	473.67 (473.09–474.59)	207.76
	binary-subspace[dimension=8] + descent-algebraic[m=2] + inherited-f4 + walk	278.79 (200.30–388.41)	122.29
	binary-subspace[dimension=8] + descent-algebraic[m=2] + inherited-f4-legacy + walk	278.79 (200.30–388.41)	122.29
	binary-subspace[dimension=8] + descent-algebraic[m=2] + fes-then-inherited + walk	278.79 (200.30–388.41)	122.29
icv1-f2m21-tm1325-b6925563 (r = 1049239)	**matched rho** negation-map r-adding walk: look-ahead, short-cycle escape by doubling, distinguished points, stride starts, 16 runs	1.24	1.00
	binary-subspace[dimension=11] + mitm[m=2] + – + walk	1975.81 (1975.78–1975.87)	1597.45
	binary-subspace[dimension=11] + descent-algebraic[m=2] + inherited-f4 + walk	4763.28 (4576.51–5019.34)	3851.13

instance (r)	configuration	S mean (min–max)	S / rho
	`binary-subspace[dimension=11] + descent-algebraic[m=2] + inherited-f4-legacy + walk`	4763.28 (4576.51–5019.34)	3851.13
	`binary-subspace[dimension=11] + descent-algebraic[m=2] + fes-then-inherited + walk`	4763.28 (4576.51–5019.34)	3851.13
`icv1-f2m13-t181-515ee569` (r = 2003)	**matched rho** `negation-map r-adding walk: look-ahead, short-cycle escape by doubling, distinguished points, stride starts`, 16 runs	2.89	1.00
	`binary-subspace[dimension=7] + mitm[m=2] + – + walk`	197.78 (197.66–198.01)	68.38
	`binary-subspace[dimension=7] + descent-algebraic[m=2] + inherited-f4 + walk`	137.95 (117.79–167.62)	47.70
	`binary-subspace[dimension=7] + descent-algebraic[m=2] + inherited-f4-legacy + walk`	137.95 (117.79–167.62)	47.70
	`binary-subspace[dimension=7] + descent-algebraic[m=2] + fes-then-inherited + walk`	101.91 (83.71–116.31)	35.24

instance (r)	configuration	S mean (min–max)	S / rho
icv1-f2m17-tm101-00378d4e (r = 65587)	**matched rho** negation-map r-adding walk: look-ahead, short-cycle escape by doubling, distinguished points, stride starts, 16 runs	1.73	1.00
	binary-subspace[dimension=9] + mitm[m=2] + – + walk	493.17 (492.55–493.54)	285.87
	binary-subspace[dimension=9] + descent-algebraic[m=2] + inherited-f4 + walk	304.50 (102.55–415.33)	176.51
	binary-subspace[dimension=9] + descent-algebraic[m=2] + inherited-f4-legacy + walk	304.50 (102.55–415.33)	176.51
	binary-subspace[dimension=9] + descent-algebraic[m=2] + fes-then-inherited + walk	304.50 (102.55–415.33)	176.51
icv1-f2m19-tm797-9c54981b (r = 262543)	**matched rho** negation-map r-adding walk: look-ahead, short-cycle escape by doubling, distinguished points, stride starts, 16 runs	1.35	1.00
	binary-subspace[dimension=10] + mitm[m=2] + – + walk	1063.26 (1062.78–1063.52)	785.16
	binary-subspace[dimension=10] + descent-algebraic[m=2] + inherited-f4 + walk	465.45 (227.52–594.50)	343.71
	binary-subspace[dimension=10] + descent-algebraic[m=2] + inherited-f4-legacy +	465.45 (227.52–	343.71

instance (r)	configuration	S mean (min–max)	S / rho
	`walk`	594.50)	
	`binary-subspace[dimension=10] + descent-algebraic[m=2] + fes-then-inherited + walk`	465.45 (227.52–594.50)	343.71
`icv1-f2m23-tm5197-1f85e9e1` (r = 4196903)	**matched rho** `signed-Frobenius r-adding walk, distinguished points (koblitz_signed_frobenius_rho_reference)`, 16 runs	0.88	1.00
	`binary-subspace[dimension=12] + mitm[m=2] + – + walk`	3986.33 (3986.00–3986.59)	4542.18
	`binary-subspace[dimension=12] + descent-algebraic[m=2] + inherited-f4 + walk`	2051.17 (1498.17–2448.60)	2337.18
	`binary-subspace[dimension=12] + descent-algebraic[m=2] + inherited-f4-legacy + walk`	2051.17 (1498.17–2448.60)	2337.18
	`binary-subspace[dimension=12] + descent-algebraic[m=2] + fes-then-inherited + walk`	2051.17 (1498.17–2448.60)	2337.18